

Course Syllabus

 Edit

Instructor

Name: Dr. David Patrick

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While I'm happy to answer questions over email, many computer science issues (especially debugging or conceptual questions) are difficult to communicate that way. If you need help, office hours are usually the fastest and least frustrating way to get help.

If you need to email me, **please include CS 2308 in the subject line**. I use an email filter that routes course emails to a folder I check more frequently, so including the course number helps me see and respond to your message faster.

I typically respond within 24 - 48 hours on weekdays. Please do not expect responses late at night, on weekends, or on holidays.

Office Hours

Hours: MW 12:30 - 2:00 and by appointment

Location: CMAL 210E

These are my official office hours, but they are not the only times you can meet with me. I

often stay later on Mondays and Wednesdays, and I'm happy to schedule additional meeting times by appointment as needed. If my office door is open (and it isn't within 15 minutes of a class time), please feel free to come in.

Office hours are for any course-related support, including debugging help, exam prep, course planning, or just talking through where you're at in the class. You do not need to be "stuck" or behind and you don't need to come with a specific question prepared (though it does help).

**Please note that I generally do not come to campus on Tuesdays or Thursdays, so in-person meetings on those days are unlikely. However, we can always schedule a virtual meeting on those days if needed.*

Teaching Assistants

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*Please allow several hours for a response and do not expect responses over weekends, holidays, or at night.

Course Format

This is a **Face to Face** course.

Section 255 Class Hours: MW 11:00 am - 12:20 pm

Location: Bruce and Gloria Ingram Hall 03104

This course uses a mix of live coding demonstrations and hands-on worksheets to teach programming concepts.

- During lectures, I will demonstrate coding concepts in real time and explain my thought process as I write and debug code.
- I will ask questions throughout class. The experience is significantly better when students participate and ask questions of their own. Don't be afraid to ask questions.
- All coding examples shown in class will be published in the zyBook after they are covered.
- You are encouraged to read the relevant zyBook sections or review completed notes **before** class.
- This format works well for most students. If you strongly prefer a different teaching style, you should consider whether this section is the best fit for you. I am happy to discuss this during office hours.

Textbook

We will be using an online textbook from zyBooks. It is **required**. We will walk through how to access and register for the zyBook during the first week of class.

Required Completion for Enrollment

You must complete the Check-in Worksheet and register for the zyBook by the roster certification date to remain enrolled in the course.

Failure to complete both items by this deadline will result in being dropped from the course.

Course Description

Fundamentals of object-oriented programming. Introduction to abstract data types (ADTs) including lists, stacks, and queues. Searching and sorting. Pointers and dynamic memory allocation. This course is a continuation of CS 1428.

Course Prerequisites

C or higher in CS 1428: Foundations of Computer Science I

Course Goals and Learning Objectives

At the end of the course, the students should be able to:

- Describe and demonstrate at least two different algorithms for searching and at least two different algorithms for sorting.
- Implement a divide-and-conquer algorithm to solve an appropriate problem (binary search).
- State the time/space efficiency of various algorithms (using one of 6 categories of mathematical functions).
- List the 6 categories of mathematical functions used in analyzing algorithms in order from slowest to fastest growing.
- Read and write C++ code that uses pointer variables and memory operations (new, &, *, delete), including pointers to arrays, structures, and objects and the -> operator.
- Write C++ code that resizes an array using dynamic memory allocation.
- Write C++ code that deletes dynamically allocated memory to avoid memory leaks.
- Describe the basic concepts of object-oriented programming.
- Design, implement, test, and debug simple programs (using objects) in an object-oriented programming language (C++).
- Describe how the class mechanism supports encapsulation and information hiding.
- Develop (implement) programs using multiple classes and arrays of objects.
- Develop and use appropriate algorithms, especially for processing lists (insert, remove, search,

sort, etc.).

- Describe structured programming in terms of modules and functions.
- Develop (implement) programs with source code separated into multiple files, including header (.h) files.
- Create, compile, and run a C++ program in a unix style command-line environment.
- Develop (Implement) C++ programs that create and use simple linked-lists, including code to insert into, delete from, and traverse a linked list structure.
- Compare and contrast the costs and benefits of dynamic and static data structure implementations.
- Describe the principle of the Abstract Data Type (ADT) and, in particular, explain the benefits of separation of interface and implementation.
- Implement user-defined data structures in a high-level language.
- Implement the list, stack, and queue ADT using arrays and linked lists.
- Write programs that use each of these data structures: linked lists, stacks, and queues.

Workload and Time Expectations

This is a work-heavy course. Learning programming requires consistent practice, and most assignments take time to think through, write, test, and debug.

Because of this, waiting until the last minute is the most common cause of problems students experience in this course. Starting assignments late often leads to incomplete work, unnecessary stress, and difficulty keeping up with the material.

You are strongly encouraged to start assignments early and seek help as soon as something feels unclear. Office hours and tutoring are most effective when used early, not right before a deadline.

Required Material

- ZyBooks

Zybooks

This interactive textbook contains readings, live coding examples from class, and most course assignments. Given the substantial number of activities throughout the semester, it is strongly recommended that you pace yourself and complete them consistently rather than attempting to

finish everything at the last minute.

Grading

The course is graded out of an 1000 point scale. Below is a breakdown of the course's point structure:

Course Point Structure

<u>Group</u>	<u>Points</u>
Check-in Worksheet	5
ZyBook Learning Activities	65
Unit Review Worksheets	90
Unit Programs	240
Exam 1	150
Exam 2	150
Final Exam	300

1. Rounding Policy:

- Final grades are determined based on the total points earned out of 1000. Fractional points will not automatically be rounded up.
- **Failing Exam Policy:** If a student fails any exam (scores below 70%), they may not qualify for any rounding considerations, regardless of their total points.
- In cases of borderline scores (e.g., 899 points), adjustments may be made at the instructor's discretion, particularly if the student's overall performance reflects consistent effort and improvement. However, this is not guaranteed.

2. Grading Scale:

- **A:** 900 or above points
- **B:** 800–899 points
- **C:** 700–799 points
- **D:** 600–699 points
- **F:** 599 or below points

3. Extra Credit Policy:

- Extra points may be made available throughout the semester through bonus activities, challenges, or participation incentives. These points can serve as "padding" to help students recover from lost points elsewhere since the grading scale remains fixed at 1000

points.

4. Flexibility of Point Structure:

- The assignment groups and their totals outlined in this syllabus are subject to change. Adjustments may be made to accommodate unforeseen circumstances, to improve the course structure, or to better reflect the workload and learning outcomes. Any changes to the assignment points or grading policy will be communicated through Canvas and during class.

Program Submission and Grading Policy

All submitted programs must compile and run to be eligible for grading. Programs that fail to compile or do not execute correctly will receive a grade of zero. It is your responsibility to thoroughly test your code to ensure it functions as expected. Partial credit will not be awarded for non-functioning programs.

If your program does not compile, office hours are the best place to get help before the deadline.

Makeup/Late Policy

Assignments and quizzes cannot be turned in late or made up without permission from the instructor.

If you need an extension due to extenuating circumstances, reach out before the deadline by email or stopping by office hours. We'll need to talk in person to work out the details. **Requests the night of or after the deadline won't be considered.**

Exam Policy

All tests will be in class during class time. You must bring your student ID to each exam (or the Texas State App). There will be no makeup tests. If you have an instructor-approved excuse, I will replace the missing test score with your final exam score.

Exam Day Procedures: Once seated for an exam, all bags, backpacks, and personal items must be placed along the sides of the classroom.

All phones must be silenced or powered off and stored in your bag. If you do not have a bag, you must turn your phone in to the front of the classroom, where it will be held until you complete your

exam.

Any phone found on your person during an exam will be treated as an academic honor violation, regardless of whether it is powered on or in use.

Attendance Policy

Programming is a skill developed through practice and feedback. Students who regularly attend office hours or ask questions early tend to perform significantly better on exams and programming assignments.

Class attendance is **highly encouraged** but it is not directly included as part of your final grade.

In-Class Response System (Top Hat)

We will use the Top Hat classroom response system during class. You will submit answers to in-class questions using a mobile device or laptop.

These questions are exam-style practice problems and will not be posted outside of class. If you miss class and want to review what was covered, come to office hours to discuss the missed material.

To register, go to the course Canvas site and click on the Top Hat link. **When prompted for a student ID, use your A0#.** Top Hat is integrated into Canvas and is free for Texas State students.

A mobile app is available in addition to the website. This is my first semester using Top Hat; if significant issues arise, we may switch platforms. If that is the case, I will announce it in class.

Student Disability Accommodations

Any student with needs requiring special accommodations should contact the office of disability services (ods.txst.edu  (<http://ods.txst.edu>)). Students who qualify for extra time for exams must take their test with ATSD and must schedule their test at the same time the test is given in class. You must submit your request with ATSD **at least 3 business days before the exam!**

Helpful Resources

Other than the instructor's office hours, there are other places to obtain assistance:

- **CS Lab tutors** (good for help on MSPs): MCS 593 (inside MCS 590 suite)
<https://cs.txstate.edu/resources/labs/tutoring/> (<https://cs.txstate.edu/resources/labs/tutoring/>)
- **CLC tutoring** (good for help on understanding the material):
<https://hlsamp.cose.txstate.edu/clc.html> (<https://hlsamp.cose.txstate.edu/clc.html>)
- **TA's office hours**: see Canvas for times and how to access.

Academic Honesty

You are expected to adhere to the University's Academic Honor Code: <https://www.txstate.edu/honorcodecouncil/Academic-Integrity.html> (<https://www.txstate.edu/honorcodecouncil/Academic-Integrity.html>)

- You may discuss problems with other students but do not share your code electronically with anyone. Do not post your code on group chats or in email!
- Do not include code obtained from the internet or any other source in your programs. Do not use features we have not covered in class without prior approval of the instructor.
- Do not collaborate with anyone during a test or exam (no talking, no cell phones or smart watches).
- The penalty for violating academic honesty on an assignment, test or exam is a reduction of points up to a **negative grade** of the assignment, test or exam.

Withdrawals/Drops

You must follow the withdrawal and drop policy set up by the University.

<https://www.registrar.txstate.edu/resources/dropping-vs-withdrawing.html> (<https://www.registrar.txstate.edu/resources/dropping-vs-withdrawing.html>)

Course Schedule

A detailed and up-to-date course calendar is maintained on the Canvas course homepage. This calendar includes lecture topics, assignment deadlines, exam dates, etc.

You are responsible for regularly checking the Canvas homepage calendar, as it is the primary source for current course scheduling information. Any major changes will also be announced in

class and via Canvas announcements.

Below is a rough outline of the course schedule.

Course Schedule

<u>Week</u>	<u>Topic</u>
0	Functions, Arrays, & Structs
1	Functions, Arrays, & Structs
2	Classes
3	Classes
4	Pointers & Dynamic Memory
5	Pointers & Dynamic Memory
6	Exam 1
7	Searching, Sorting, & Analysis
8	Spring Break (No classes)
9	Searching, Sorting, & Analysis
10	Linked Lists
11	Linked Lists and Stacks & Queues
12	Stacks & Queues
13	Exam 2
14	Exam Review
15	Final Exam Prep

Minor changes to the course schedule may occur throughout the semester to better align with the course content and objectives. For a list of important university-wide dates, review [Office of the University Registrar Academic Calendar](https://www.registrar.txst.edu/registration/ac/academic-calendar.html)  (<https://www.registrar.txst.edu/registration/ac/academic-calendar.html>)

Exam Topics

Exam 1 will cover Units 1, 2, and 3.

Exam 2 will cover Units 4, 5, and 6.

Final exam will be a comprehensive exam of Units 1 - 6.